Notes: Kyle (Econometrica, 1985)

Continuous Auctions and Insider Trading

Contents

1 Big picture

- **Question.** How quickly is private information incorporated into prices when an insider trades strategically over time? What determines market depth, price volatility, and the value of private information?
- **Setting.** The paper studies a speculative market with **one risky asset**, **one monopolistic informed trader**, **random noise traders**, and **competitive market makers**. Trading first takes place in a sequence of auctions and then in the limit of **continuous trading**.
- **Core challenge.** Prices are assumed to be efficient in the semi-strong sense, but the insider is strategic and internalizes price impact. The paper therefore needs an equilibrium in which **market efficiency and strategic trading coexist**.
- **Main theoretical contribution.** The paper characterizes a **unique linear equilibrium** in:
  - a single-auction model,
  - a sequential-auction model,
  - and a limiting continuous-auction model.

  It shows exactly how order flow reveals information, how market depth depends on noise trading and remaining private information, and how the discrete model converges to the continuous limit.
- **Main mechanism.** The insider behaves like an **intertemporal monopolist**. He does *not* reveal all information immediately, because aggressive current trading moves prices against him and destroys future rents. Noise traders provide camouflage, allowing him to trade gradually.
- **Why it matters.** This paper is one of the foundational microstructure models behind the modern notion of **Kyle's** λ, i.e. the price impact of order flow. It also gives a rigorous theory of **market liquidity** as an endogenous consequence of asymmetric information.
- **Main results.**
  - In the **single-auction equilibrium**, the insider trades linearly on his information, market makers price linearly on total order flow, and **one-half** of the insider's private information is incorporated into price immediately.
  - In the **sequential-auction equilibrium**, information is incorporated **gradually**; the insider chooses how fast to reveal information depending on current and future market depth.
  - With finitely many auctions, the residual variance of private information, denoted Σ_n, declines over time but remains **strictly positive** at the end of trading.
  - In the **continuous-auction limit**, prices follow **Brownian motion**, market depth is **constant over time**, and **all private information is incorporated into prices by the end of trading**.
  - More noise trading increases market depth proportionately and increases insider profits proportionately, but it does **not** change the informativeness of prices in the basic linear-normal environment.
  - The expected profits of the insider in the continuous limit are **twice** those in the single-auction model.

- **Relation to earlier intuition.** The paper makes precise Bagehot's idea that adverse selection makes markets less liquid and connects it to Black's notions of **tightness**, **depth**, and **resiliency**.
- **Economic takeaway.** A liquid market is not just a place with many trades. In this paper, liquidity is an equilibrium object determined by the interaction of **private information**, **strategic insider trading**, and **noise trading**. Strategic insiders reveal information smoothly because immediate revelation is too expensive.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no empirical dataset**. The central objects are the primitives of the trading game:
  - the liquidation value of the risky asset,
  - the insider's order,
  - noise-trader order flow,
  - the price set by market makers,
  - and the amount of private information not yet incorporated into prices.
- The paper proceeds in three layers:
  - a **single-auction model**,
  - a **sequential-auction model**,
  - a **continuous-auction limit**.

2.2 Agents and their information

- **Insider.**
  - risk neutral,
  - observes the liquidation value v privately,
  - chooses order quantities strategically.
- **Noise traders.**
  - uninformed,
  - submit random orders,
  - create camouflage for the insider.
- **Market makers.**
  - competitive and risk neutral,
  - do not see insider and noise orders separately,
  - only observe **aggregate order flow**,
  - set prices equal to conditional expectations, so they earn zero expected profits.

2.3 Asset value, noise trading, and order flow

- In the **single-auction model**,

$$v \sim N(p_0, \Sigma_0), \qquad u \sim N(0, \sigma_u^2),$$

  independently.

- The insider submits order x and the market makers observe only total order flow

$$y = x + u.$$

- In the **sequential model**, trading takes place on a grid

$$0 = t_0 < t_1 < \cdots < t_N = 1,$$

  and noise trading is generated by a Brownian motion $u(t)$ with instantaneous variance σ_u^2, so that

$$\Delta u_n \sim N(0, \sigma_u^2 \Delta t_n), \qquad \Delta t_n = t_n - t_{n-1}.$$

2.4 Timing of trade

- At each auction, trading occurs in **two steps**:
  1. the insider and the noise traders choose quantities,
  2. market makers observe only total order flow and set the clearing price.
- In the continuous-time discussion, the same logic is maintained through **end-of-instant pricing**: the insider's current order affects the current execution price.
- This timing is essential. If trades were priced before the insider's own order affected the quote, the insider would want to trade unbounded quantities immediately.

2.5 Key endogenous objects

- p: the price in the single-auction model.
- p_n: the price after auction n in the sequential model.
- λ or λ_n: the **liquidity parameter**, i.e. the slope of price with respect to order flow. Small λ means a **deep** market.
- β or β_n: the insider's trading intensity on the basis of private information.
- Σ_n: the error variance of price after auction n, i.e. how much private information remains hidden from market makers.
- α_n, δ_n: parameters describing the insider's continuation value in the dynamic program.

2.6 Liquidity concepts emphasized by the paper

- **Tightness:** the cost of turning around a position over a short interval.
- **Depth:** the amount of order flow needed to move price by one unit; measured by $1/\lambda$.
- **Resiliency:** the speed with which prices return toward underlying value after an uninformative shock.

Interpretation. The paper's key innovation is that these liquidity concepts are not imposed externally. They are derived from optimal insider behavior and competitive pricing under asymmetric information.

3 Models and empirical specifications (all equations + interpretation)

3.1 Single-auction equilibrium: environment and equilibrium concept

The insider's profits are

$$\pi = (v - p)x.$$

An equilibrium is a pair of functions (X, P) satisfying:

$$\mathbb{E}[\pi(X, P) \mid v] \geq \mathbb{E}[\pi(X', P) \mid v]$$

for every alternative insider strategy X', and

$$p(X, P) = \mathbb{E}[v \mid x + u].$$

Interpretation.

- The first condition says the insider optimally chooses quantity, taking the pricing rule as given.
- The second condition says market makers price efficiently from aggregate order flow alone.
- This is effectively the reduced form of a Bertrand competition among market makers, which drives their expected profits to zero.

3.2 Single-auction linear equilibrium

Kyle shows there is a unique linear equilibrium of the form

$$x = \beta(v - p_0), \qquad p = p_0 + \lambda(x + u),$$

with

$$\beta = \left(\frac{\sigma_u^2}{\Sigma_0}\right)^{1/2}, \qquad \lambda = \frac{1}{2}\left(\frac{\Sigma_0}{\sigma_u^2}\right)^{1/2}.$$

The price informativeness after the single auction is

$$\Sigma_1 \equiv \mathrm{Var}(v \mid p) = \frac{1}{2}\Sigma_0.$$

Expected ex ante profits are

$$\mathbb{E}[\pi] = \frac{1}{2}\left(\Sigma_0 \sigma_u^2\right)^{1/2}.$$

Interpretation.

- The insider trades more aggressively when noise trading is large because more camouflage is available.
- Market makers respond to stronger informational asymmetry with a steeper price impact coefficient λ.
- Exactly one-half of the insider's information is revealed in one shot: the market learns something from order flow, but not everything.
- The object $1/\lambda$ is market depth. More noise trading makes the market deeper, because any given order-flow realization is less informative.

3.3 Sequential-auction equilibrium: dynamic setup

Trading occurs at auction dates $t_1, \ldots, t_N$. Let Δx_n be the insider's trade at auction n, Δu_n the noise-trader order, and Δp_n the price change.

A sequential auction equilibrium consists of strategies and pricing rules such that

$$p_n = \mathbb{E}[v \mid \Delta x_1 + \Delta u_1, \ldots, \Delta x_n + \Delta u_n],$$

and, at each auction n, the insider's current order maximizes expected profits from rounds $n, \ldots, N$ given current information.

Interpretation.

- The insider cannot commit to a future trading path.
- He therefore chooses each trade to maximize continuation profits, knowing that aggressive current trading reveals information and affects later prices.
- Prices form a martingale because market makers always condition on the full order-flow history they observe.

3.4 Recursive linear equilibrium in the sequential model

Kyle proves there is a unique recursive linear equilibrium in which

$$\Delta x_n = \beta_n (v - p_{n-1}) \Delta t_n, \tag{3.11}$$

$$\Delta p_n = \lambda_n (\Delta x_n + \Delta u_n), \tag{3.12}$$

$$\Sigma_n = \mathrm{Var}(v \mid \Delta x_1 + \Delta u_1, \ldots, \Delta x_n + \Delta u_n), \tag{3.13}$$

and the insider's continuation value satisfies

$$\mathbb{E}[\pi_n \mid p_1, \ldots, p_{n-1}, v] = \alpha_{n-1}(v - p_{n-1})^2 + \delta_{n-1}. \tag{3.14}$$

The equilibrium coefficients solve the system

$$\alpha_{n-1} = \frac{1}{4\lambda_n(1-\alpha_n\lambda_n)}, \tag{3.15}$$

$$\delta_{n-1} = \delta_n + \alpha_n\lambda_n^2\sigma_u^2\Delta t_n, \tag{3.16}$$

$$\beta_n\Delta t_n = \frac{1-2\alpha_n\lambda_n}{2\lambda_n(1-\alpha_n\lambda_n)}, \tag{3.17}$$

$$\lambda_n = \frac{\beta_n\Sigma_n}{\sigma_u^2}, \tag{3.18}$$

$$\Sigma_n = (1-\beta_n\lambda_n\Delta t_n)\Sigma_{n-1}, \tag{3.19}$$

subject to

$$\alpha_N = \delta_N = 0$$

and the second-order condition

$$\lambda_n(1-\alpha_n\lambda_n) > 0. \tag{3.20}$$

Using market efficiency, one can also write the pricing and information-update equations as

$$\lambda_n = \frac{\beta_n\Sigma_{n-1}}{\beta_n^2\Sigma_{n-1}\Delta t_n + \sigma_u^2}, \qquad \Sigma_n = \frac{\sigma_u^2\Sigma_{n-1}}{\beta_n^2\Sigma_{n-1}\Delta t_n + \sigma_u^2}. \tag{3.29}$$

Interpretation.

- β_n measures how aggressively the insider trades on the remaining mispricing $v - p_{n-1}$.
- λ_n measures contemporaneous price impact: large λ_n means a thin market, small λ_n means a deep market.
- Σ_n is the amount of private information still hidden from market makers after auction n.
- α_n summarizes the value of future trading opportunities; when future opportunities are valuable, the insider trades more cautiously today.

3.5 What the sequential system means economically

The recursive system implies:

- **Gradual information revelation.** Since Σ_n declines over time, more and more information is built into prices, but with finitely many auctions some information typically remains hidden at the end.
- **Strategic smoothing.** If future market depth is expected to be greater than current depth, the insider prefers to **save information** and trade more later. If future depth is expected to be worse, he trades more intensely now.
- **Noise trading as camouflage.** If σ_u doubles, the market becomes deeper, the insider trades more, and his profits rise proportionately, while the informativeness path Σ_n is unaffected.
- **Bagehot's intuition.** Market makers protect themselves against adverse selection by making the market less liquid when the insider has more information or trades more aggressively.

3.6 Continuous-auction equilibrium: trading and pricing rules

In the continuous limit, Kyle studies trading rules of the form

$$\mathrm{d}\pi(t) = [v - p(t)]\,\mathrm{d}x(t), \tag{4.1}$$

$$\mathrm{d}x(t) = \beta(t)[v - p(t)]\,\mathrm{d}t, \tag{4.2}$$

$$\mathrm{d}p(t) = \lambda(t)\big[\,\mathrm{d}x(t) + \mathrm{d}u(t)\big]. \tag{4.3}$$

The market efficiency condition is

$$p(t) = \mathbb{E}\big[v \,\big|\, \{\,\mathrm{d}x(s) + \mathrm{d}u(s)\}_{s\in[0,t]}\big]\,. \tag{4.5}$$

Define

$$\Sigma^*(t) = \mathbb{E}[(v - p(t))^2]. \tag{4.6}$$

$$\Sigma(t) = \mathrm{Var}\big(v \,\big|\, \{\,\mathrm{d}x(s) + \mathrm{d}u(s)\}_{s\in[0,t]}\big)\,. \tag{4.7}$$

Market efficiency implies $\Sigma^*(t) = \Sigma(t)$.

Interpretation.

- The insider buys or sells smoothly in response to the remaining gap between value and price.
- End-of-instant pricing guarantees that current price impact is internalized by the insider.
- The state variable $\Sigma(t)$ is the remaining private information still not priced in.

3.7 Continuous-auction equilibrium: closed-form solution

Kyle's main continuous-time theorem gives

$$\lambda(t) = \left(\frac{\Sigma_0}{\sigma_u^2}\right)^{1/2}, \tag{4.8}$$

$$\Sigma(t) = (1-t)\Sigma_0, \tag{4.9}$$

$$\beta(t) = \frac{\sigma_u^2\lambda(t)}{\Sigma(t)} = \frac{\sigma_u\Sigma_0^{-1/2}}{1-t}, \tag{4.10}$$

$$\alpha(t) = \frac{1}{2}\left(\frac{\sigma_u^2}{\Sigma_0}\right)^{1/2}, \qquad t \in (0,1), \tag{4.11}$$

$$\delta(t) = \frac{1}{2}\big(\sigma_u^2\Sigma_0\big)^{1/2}(1-t). \tag{4.12}$$

Because

$$\lambda(t)^2\sigma_u^2 = \Sigma_0,$$

the price process has constant instantaneous variance Σ_0.

The insider's ex ante profits in the continuous limit are

$$\mathbb{E}[\pi(0)] = \alpha(0)\Sigma_0 + \delta(0) = \big(\Sigma_0\sigma_u^2\big)^{1/2},$$

which is exactly double the single-auction value.

Interpretation.

- **Constant depth.** The price impact parameter $\lambda(t)$ is constant over time, so market depth $1/\lambda(t)$ is constant.
- **Constant volatility.** Since noise trading drives the instantaneous variance of prices, prices look like Brownian motion even though they gradually reveal private information.
- **Linear information decay.** Remaining private information falls at the linear rate $\Sigma(t) = (1-t)\Sigma_0$.
- **Increasing resiliency.** Because $\beta(t) \to \infty$ as $t \to 1$, the insider trades increasingly aggressively near the end; prices become infinitely resilient as the terminal date approaches.
- **Full revelation by the end.** Since $\Sigma(1) = 0$, price converges to value by the end of trading.

3.8 Why constant depth emerges in continuous time

Kyle transforms the insider's optimization problem into one over the state $\Sigma^*(t)$ and obtains

$$\mathbb{E}[\pi(0)] = \frac{1}{2}\int_0^1 \lambda(t)\sigma_u^2 \,\mathrm{d}t + \frac{1}{2}\int_0^1 \lambda(t)^{-1} \,\mathrm{d}[-\Sigma^*(t)]. \tag{4.19}$$

Interpretation.

- If $\lambda(t)$ were ever to **decrease** later on, the insider could destabilize prices before the decrease and then profit from unwinding at better terms.
- If $\lambda(t)$ were ever to **increase** later on, the insider would want to reveal all information immediately before the market becomes less deep.
- Therefore any time variation in depth creates profitable deviations. The only sustainable continuous-time equilibrium has **constant** $\lambda(t)$.

3.9 Convergence from discrete to continuous trading

Let $|\Delta t|$ denote the mesh of the trading partition. Kyle proves that as $|\Delta t| \to 0$, the sequential-auction coefficients converge to the continuous-auction objects.

In particular, the piecewise-constant interpolants of

$$\lambda_n, \beta_n, \Sigma_n, \alpha_n, \delta_n$$

converge to the continuous-time formulas for

$$\lambda(t), \beta(t), \Sigma(t), \alpha(t), \delta(t),$$

with uniform convergence for $\Sigma(t)$ and $\delta(t)$ on $[0,1]$ and uniform convergence for $\lambda(t)$, $\beta(t)$, and $\alpha(t)$ away from $t = 1$.

Interpretation.

- The continuous model is not a separate heuristic story. It is the **limit** of the finite-auction model.
- As auctions become more frequent, the insider reveals information more smoothly, the discrete liquidity schedule flattens, and the residual information path converges to the linear continuous-time path.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a theory paper: what pins down the equilibrium

- There is no empirical identification problem here. The relevant question is what pins down the equilibrium **analytically**.
- The equilibrium is disciplined by three ingredients:
  - **profit maximization** by the insider,
  - **market efficiency** by market makers,
  - **normality plus linearity**, which make conditional expectations tractable.
- Together these ingredients yield a **unique linear equilibrium** in the single-auction model and a **unique recursive linear equilibrium** in the sequential model.

4.2 Why the insider trades gradually

- The insider is not a price taker. He understands that aggressive current trading moves the price and reveals information.
- Trading too much today therefore has two costs:
  - it worsens the current execution price,
  - and it reduces future rents by revealing information.
- This is why the insider spreads trades over time instead of dumping all information into price immediately.

4.3 Why noise trading matters

- Noise trading is not just a technical device. It is what makes insider profits possible.
- Without noise traders, market makers could infer the insider's information too quickly and the market would become too thin for profitable insider trading.
- More noise trading increases camouflage, deepens the market, and raises insider profits.

4.4 Why the second-order condition matters

- The discrete-time second-order condition

  $$\lambda_n(1 - \alpha_n\lambda_n) > 0$$

  rules out destabilization strategies in which the insider intentionally pushes the price away from value today to profit more tomorrow.
- Economically, it limits the insider's ability to exploit future trading opportunities by first creating artificial mispricing.
- In continuous time, the same logic becomes the argument that **time-varying depth cannot be sustained**: if depth changed over time, the insider would exploit that variation.

4.5 Why the market-efficiency assumption is powerful

- Market makers price only from order flow, so all price changes are ultimately consequences of order-flow innovations.
- Yet prices are still efficient because market makers use the conditional expectation of value given the order flow they observe.
- This is one of the paper's deepest points: **efficient prices do not require competitive informed trading**. Prices can be efficient even when a monopolistic insider strategically smooths information revelation.

4.6 What assumptions do the heavy lifting

- a **single** informed trader,
- **risk neutrality** of insider and market makers,
- **competitive** market making,
- **normally distributed** fundamentals and noise orders,
- **market orders** rather than limit orders,
- exogenous noise trading and no inventory costs.

These assumptions make the model elegant and tractable, but they also mean the paper abstracts from many features that later market microstructure models study explicitly.

4.7 Relation to the literature

- Relative to **Bagehot**, the paper formalizes the idea that adverse selection makes markets less liquid.
- Relative to **Black**, it provides an endogenous model of tightness, depth, and resiliency.
- Relative to **Glosten–Milgrom**, it emphasizes a **strategic monopolistic insider** and gradual information revelation in a dynamic setting.
- The paper became the classic source of the idea that order-flow price impact can be summarized by **Kyle's** λ.

5 Tables and figures

5.1 No empirical tables

- This paper is entirely theoretical. There are no regression tables or estimation results.
- The key quantitative object is the figure that compares the discrete and continuous equilibrium paths.

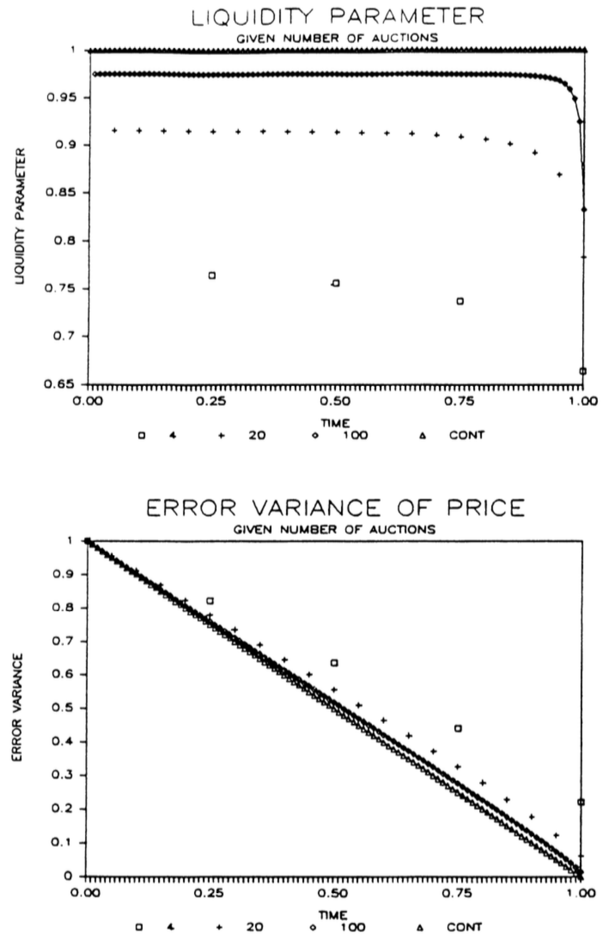

FIGURE 1

5.2 Figure 1: liquidity parameter and error variance as the number of auctions increases

What is plotted.

- Top panel: the liquidity parameter λ_n for $N = 4$, $N = 20$, $N = 100$, and the continuous limit.
- Bottom panel: the error variance Σ_n under the same cases.
- The normalization used is $\Sigma_0 = \sigma_u^2 = 1$, with equally spaced auctions.

How to read it.

- In the continuous model, the benchmark is

$$\lambda(t) = 1, \qquad \Sigma(t) = 1 - t.$$

- For finite N, the discrete paths differ from these limits because information is revealed in blocks rather than continuously.

Main takeaway.

- As N becomes larger, the path of λ_n becomes flatter and moves toward the constant continuous-time value.
- The residual-information path Σ_n gets closer and closer to the straight line $1 - t$.
- With only a few auctions, substantial private information remains hidden until late in the trading day. With many auctions, revelation becomes almost continuous.

Economic meaning. Figure 1 is the paper's visual proof that the continuous-auction model is not merely heuristic. It is the limiting object toward which the discrete model converges.

6 Short summary

- **Method.** A dynamic linear-normal model of insider trading with competitive market makers and random noise traders.
- **Core mechanism.** The insider trades gradually because current aggressive trading raises price impact and reduces future informational rents.
- **Main equilibrium result.** In continuous time, prices follow Brownian motion, market depth is constant, and all private information is incorporated into prices by the end of trading.
- **Main lesson.** Liquidity is an endogenous outcome of asymmetric information and strategic trading, not an exogenous market characteristic.

7 Conclusion

7.1 Core conclusion

- The paper shows that a market can be efficient even when prices are shaped by the strategic behavior of a monopolistic insider.
- In equilibrium, information is revealed through order flow rather than arriving directly into prices.
- The sequential-auction model converges cleanly to a continuous-auction equilibrium with constant depth and full revelation at the end of trading.

7.2 Economic intuition

- The insider wants to exploit private information, but exploiting it too quickly reveals it.
- Noise traders hide the insider's trades, allowing him to trade more without moving prices too much.
- Market makers react to the informativeness of order flow by adjusting price impact.
- The resulting interaction generates equilibrium liquidity, volatility, and information revelation endogenously.

7.3 Takeaway

This paper is foundational because it turns vague ideas about insider trading and liquidity into a fully worked-out equilibrium theory. Its lasting lesson is that order flow is informative, price impact is endogenous, and the speed of information revelation reflects the strategic trade-off between exploiting information today and preserving it for tomorrow.